Nowcasting of Storms Using Predicted Integrated Water Vapor With a Machine Learning Technique and Satellite Brightness Temperature

Deepak S. Bisht, T. Narayana Rao, N. Rama Rao, *Senior Member, IEEE*, and S. V. Chandrakanth

***Abstract*— A hybrid model for nowcasting of storms has been developed by employing predicted integrated water vapor (IWV) with light gradient boosting machine (GBM) on Global Navigation Satellite System (GNSS) receiver data collected at Gadanki (13.46°N, 79.17°E) and estimated thresholds for three storm predictors. The utilization of predicted IWV allows more lead time for disaster preparedness. The efficacy of light GBM technique in predicting IWV has been tested on 54 stormy days (from the year 2020), identified with a collocated polarimetric weather radar observations. The predicted IWV agrees very well with observed IWV with rms errors < 1.2 mm (correlation coefficient > 0.85) for predictions with a lead time up to 2 h. Among several predictors considered for nowcasting, IWV is found to have a great predictive potential, as the moisture buildup is seen few hours (1–4 h) prior to the occurrence of storm/rainfall. Thresholds for chosen predictors [magnitude of IWV, change in IWV, and change in brightness temperature (T_b)] are finalized using the data from known stormy days (65 days from the years 2018 and 2019). The sensitivity analysis of the predictors independently and in combination in predicting storms reveals that 1 and 2 parameter-based predictions detect storms accurately but produce large false alarm rates. The three-parameter scheme reduced the false alarm rate drastically to 5% and improved the model accuracy to 97%, which is much better than the existing methods.**

***Index Terms*— Global Navigation Satellite System (GNSS) receiver, integrated water vapor (IWV), machine learning technique, nowcasting of storms.**

I. Introduction

STORMS are life-threatening weather events with huge damage potential to property as they are accompanied by intense rain, flash flooding, hail, lightning, violent winds, and even tornadoes. They occur at various scales, from mesoscale thunderstorms to synoptic scale cyclones. The prediction of these storms is of utmost importance, and their prediction capability has increased tremendously in recent years with the advances in numerical weather prediction (NWP) models. These advances include both computational facilities and improvements in model physics and initial conditions. Although the prediction by NWP models is found to be reliable for large-scale systems, they often fail to predict small-scale systems, such as localized storms. These systems, though localized, often create havoc due to intense rain leading to flash floods in mountainous regions and traffic and water logging in urban regions and most importantly produce deadly lightning. Several studies have now shown that the fatalities due to lightning have increased in recent years, and in fact, they are estimated to be more than that of cyclones. These storms are also of major concern for aviation safety and air traffic management [1]. Recent studies have shown that these storms can be nowcasted (1–6 h in advance) with reasonable accuracy with the help of rapidly updated high temporal resolution datasets obtained with automatic weather stations (AWSs) [2], Doppler weather radars, satellites (cloud maps with infrared channels), lightning networks, and multiwavelength radiometers [3] Global Navigational Satellite Systems (GNSSs) networks [4], [5], [6]. In recent years, blending techniques employing different datasets are also being developed for improved nowcasting [1]. Nowcasting significantly reduces fatalities and property damage and also plays a key role in crisis management [7].

The GNSS-based nowcasting techniques are increasingly becoming popular due to the availability of high resolution, low latency, and round-the-clock data of integrated water vapor (IWV) [4], [5], [6], [8], [9]. Furthermore, the GNSS receiver networks are being established at a rapid phase in all the countries, providing a unique opportunity to map water vapor and its variability on a global scale. It is known that water vapor is the basic ingredient for convective storms/rain [10], but their relation is complex and varies from land to ocean and also between different oceanic regions [11], [12]. At a local scale, moisture builds up rapidly before the occurrence of rainfall/storm and this signature has been used in several studies to nowcast rainfall/storm [1], [4], [5], [6], [13], [14], [15].

The GNSS-based nowcasting models typically use a threshold magnitude and/or gradient of IWV for identifying storm/rain occurrence [4], [5], [6]. Some of the studies used

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hybrid models, wherein GNSS data are coupled with meteorological data or radar observations or lightning data to predict the storm [1], [4], [15] noted a positive correlation between the large gradients in IWV and rainfall occurrence and therefore devised a method to nowcast rainfall based on IWV increment. Although the model's predictability is good (75%), false alarm is also high at 40%–65%, depending on the threshold IWV gradient. Yao et al. [5] used three parameters of IWV: magnitude, variation, and rate, as indicators for nowcasting rainfall. The inclusion of other IWV parameters has increased the detection rate by 7%; however, the false alarm rate is nearly remained the same (66%). Subsequent studies also used IWV increment and IWV slope [16] or magnitude of IWV and rate of IWV [6] or magnitude of IWV, IWV increment, and rate of IWV [17] with different thresholds to improve the predictability of storms/rainfall and also to reduce the false alarm rate. Zhao et al. [14] improved the rainfall forecasting model with the help of five predictors, monthly IWV, seasonal IWV, ZTD fluctuations, and their first derivatives. The model, indeed, increased the detection rate to 95% and reduced the false alarm rate to less than 30%. To reduce the false alarm rate, researchers focused on blending techniques by employing data from a variety of instruments, such as AWS, GNSS receivers, infrared channels of satellites, and DWRs. The model developed by Benevides et al. [18] employing satellite-derived cloud top temperature and pressure along with IWV, air pressure, relative humidity, surface air pressure, and temperature to predict rainfall events had a success rate of 64% and a false alarm rate of ∼22%. Guerova et al. [15] developed a "convective storm demonstrator" using IWV and instability indices. The implementation of this algorithm on the data collected at 12 locations in Bulgaria reveals that the predictability (false alarm rate) of the model is 83% (38%). It can be inferred from the above studies that IWV-derived parameters can be used as primary predictors of storm events, while a well-chosen secondary parameter reduces the false alarms.

Recently, artificial intelligence and machine learning (AI/ML) techniques have become popular in geosciences, including GNSS-based prediction of IWV [19], [20], [21] and nowcasting applications [1], [22], [23], [24], [25]. Some of these studies used zenith tropospheric delay (ZTD) to predict extreme events [24], [25]. The usage of predicted IWV allows nowcasting of storms/rainfall in advance, providing more time for disaster managers.

As seen above, some of the studies are qualitative and others are based on simple gradient methods quantifying the changes in water vapor. Recent studies use advanced artificial intelligence techniques and predict storms/rain. However, the nowcasting of storms by the above approaches provides a very short lead time for disaster preparedness, 5–30 min. Furthermore, false alarm is found to be large with all models/algorithms. In the present study, a hybrid model is developed to nowcast storms wih different lead times (up to 2 h) by blending predicted IWV by employing an AI/ML technique with GNSS receiver and AWS data as inputs with satellite measurements. This model combines the strengths of earlier techniques and provides a robust model to improve the detection capability of storms with reduced false alarms. Furthermore, early prediction of storms will provide more lead time for disaster preparedness.

II. Data

An integrated system, consisting of a GNSS receiver (Trimble NetR9) and AWS (Vaisala WXT536), has been established at Gadanki (13.46°N, 79.17°E) for continuous monitoring of IWV. This system is a part of seven such receivers installed within 6–8 km around Gadanki and being operated in a network mode to retrieve the profiles of water vapor employing tomographic inversion techniques. Gadanki is located in southeastern India and is influenced by both southwest and northeast monsoon seasons [26]. Most of the rainfall occurs in the form of convective rain during premonsoon and monsoon seasons associated with isolated thunderstorms and mesoscale convective systems [26], [27].

The GNSS receiver measurements obtained during September 2018–December 2020 have been used for the present study. In addition, other meteorological parameters, such as surface temperature, surface pressure, wind speed and direction, and accumulated rainfall obtained from AWS, are used. Near real-time brightness temperature (T_b) retrieved from Indian National Satellite (INSAT-3DR) at 30-min resolution is also used to monitor cloud growth and decay. An X-band dual-polarization Doppler weather radar has been used to detect the storms and their movement in and around Gadanki. The radar is capable of providing such information up to a range of 80 km.

A. Retrieval of GPS-Derived IWV

As GNSS signals travel from the satellite to the GNSS receiver on the ground, the atmosphere and ionosphere delay their arrival. The ionosphere being a dispersive medium delays different frequencies differently. This delay can be estimated with multifrequency (in our case, dual frequency) GNSS receiver and can be removed from slant tropospheric delay to obtain the tropospheric delay. The slant delays in different directions are mapped to the zenith using the Vienna mapping function to obtain ZTD. The GNSS receiver data sampled at 1 Hz were postprocessed using GAMIT/GLOBK software with a sample rate of 30 min and an elevation cutoff angle of 15°. The zenith hydrostatic delay (ZHD) and the zenith wet delay (ZWD) are the two components of the ZTD [28]. The ZHD is calculated using the surface pressure measured with the AWS, which is then subtracted from the ZTD to obtain ZWD [8]. The ZWD is converted into IWV using the following equations [8]:

$$\mathrm{IWV} = \mathrm{ZWD} * \pi \tag{1}$$

$$\pi = 106/(\rho_w \times R_v \times (k_3/T_m + k_2 - k_1 \times m_w/m_d)) \tag{2}$$

where ρ_w is the density of liquid water ($= 1 \times 10^3$ kg·m^{-3}), R_v is the gas constant for water vapor ($=$ 461.524 J·kg^{-1}·K^{-1}), T_m is the atmospheric weighted mean temperature (K), k_1–k_3 are the atmospheric refraction coefficients, m_w is the molar mass of vapor, and m_d is the molar mass of dry air. T_m is estimated from the surface temperature

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(T_s), using an empirical formula derived for Gadanki using five years of radiosonde measurements

$$T_m = 70.2 + 0.72T_s. \quad (3)$$

III. Methodology and Results

A. Machine Learning Technique (Light GBM)

The nowcasting technique proposed in the present study is a two-step process: 1) prediction of IWV using an ML technique (light GBM), following Bisht et al. [21] and 2) fixing thresholds to chosen predictors for nowcasting storms. The gradient boosting machine (GBM) can be utilized for nowcasting by its ability to quickly learn from historical data and make predictions in near real time. Since GBM is capable of handling large volumes of data and can adapt to changing patterns, it can effectively capture short-term trends and patterns present in the data for nowcasting purposes. Recently, Bisht et al. [21] employed LightGBM, a machine learning algorithm developed by Microsoft, which is built upon the Gauss one side sampling approach, enhancing its efficiency and performance in various machine learning tasks for predicting IWV. The predicted IWV with different lead times was tested for several days (one year), including storm days. The algorithm was able to predict IWV with good accuracy up to 2 h with seasonal root mean square values $\leq$ 2 mm.

The validity of the algorithm has been tested here for various storm cases [two isolated thunderstorm events (July 15 and August 7, 2020) occurring at different times of the day and one long event (July 8, 2020) with multiple cells of convection] at different lead times (30–120 min) in Fig. 1. The increase in IWV is seen in almost all events, while it is concomitant with the rainfall in some cases and observed much before the event in other cases. The predicted IWV could accurately reproduce the observed temporal variability of IWV for all three events and for all lead times. The model can predict even the storm-associated increase in IWV accurately. Therefore, the predictive model will be quite effective in nowcasting the storms, as it is possible to obtain IWV and monitor its variation before the storm occurrence. The root mean square error (RMSE) in forecast IWV for all lead times is $<$ 2 mm. The model prediction of IWV is compared with the observed IWV for all the storm cases, and the statistics of correlation analysis for different lead times are shown in Table I. A very good correlation is found with a correlation coefficient $\geq$ 0.85 and RMSE $<$ 1.2 mm up to 120-min prediction.

B. Determining Thresholds

To identify storms unambiguously with an automated algorithm using moisture buildup as a criterion, several storms need to be examined to arrive at thresholds for predictor(s). For this purpose, storms passing close to Gadanki location (within 1 km) have been detected using measurements from the X-band dual-polarization radar at Gadanki.

Fig. 2 shows a typical example of storm occurrence (on July 13, 2019) as seen by the X-band radar, AWS,

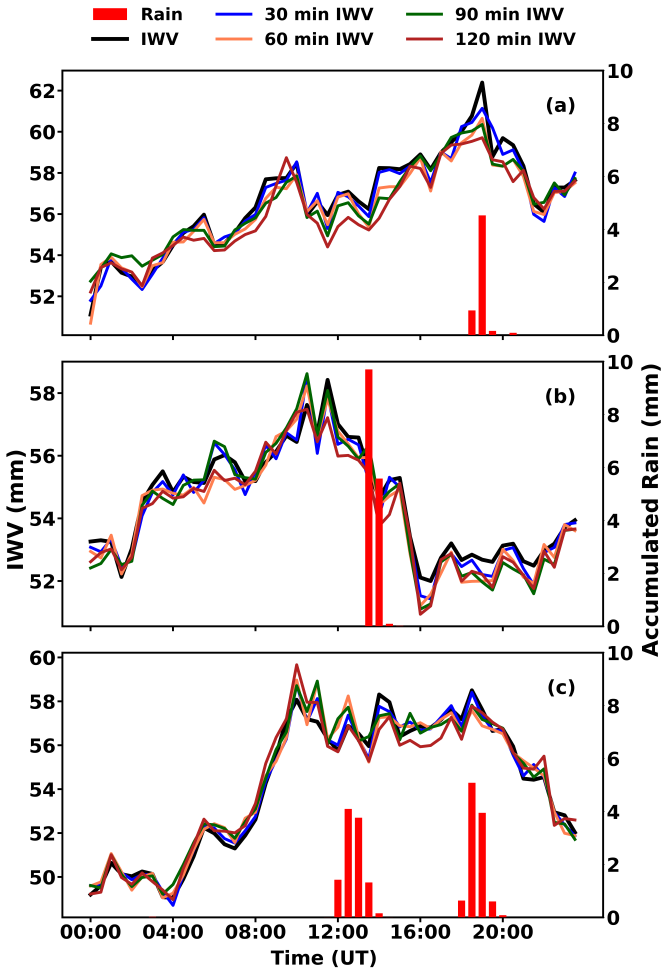


Fig. 1. Predicted IWV with light GBM technique along with GNSS-derived IWV and rainfall on three storm days. (a) July 15, 2020, (b) August 7, 2020, and (c) July 8, 2020.

TABLE I

Statistics of Correlation Analysis for Storm Days, Depicting the Accuracy of Model's Prediction of IWV for Different Leadtimes. R^2 and MAE Indicate Correlation Coefficient and Mean Absolute Error, Respectively

Statistical metric	30 min	60 min	90 min	120 min
R^2	0.96	0.93	0.90	0.85
RMSE	0.42	0.72	0.91	1.17
MAE	0.36	0.50	0.66	0.84

INSAT-derived T_b, and GNSS receiver. The spatial variation of reflectivity factor (Z) at horizontal polarization at 8° elevation around Gadanki (10 $\times$ 10 km), as observed by the radar [see Fig. 2(a)], clearly shows convective cells ($Z \geq$ 40 dBZ) embedded in a large precipitating system. The cells on the southwest side passed over the study site and produced a good amount of rainfall at Gadanki. The AWS measurements clearly show copious rainfall (about 45 mm in 5 h) starting from 15:30 UTC at Gadanki. The AWS measurements logged at 1-s resolution are averaged over 30 min (accumulated for rainfall) to match with GNSS receiver measurements. Sudden drop in temperature and rapid increase in wind speed can also

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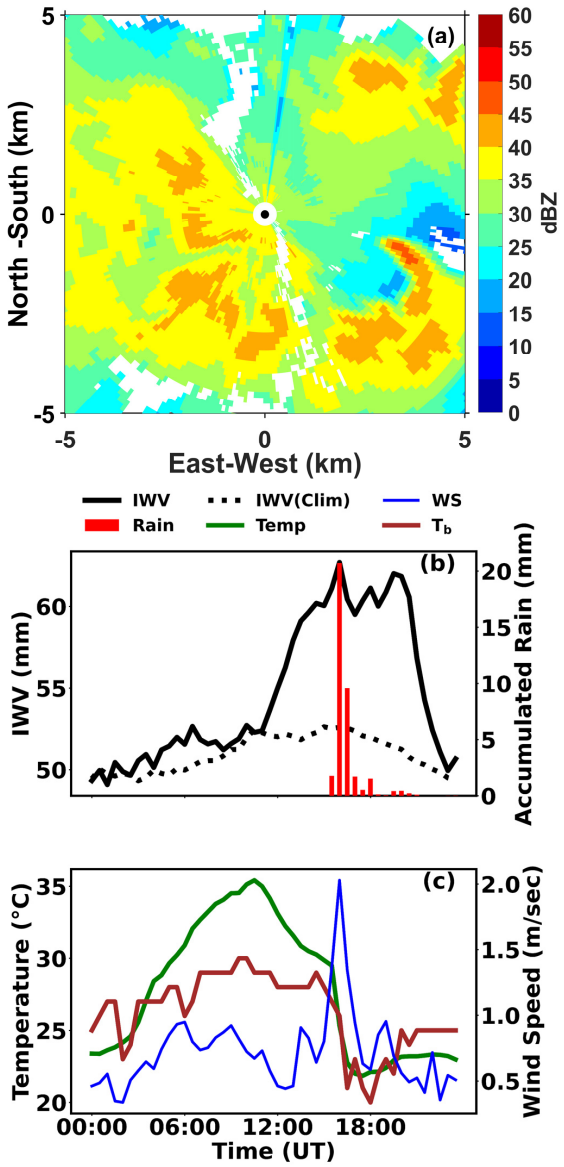


Fig. 2. (a) Plane position indicator (PPI) for Z at 8° elevation obtained with an X-band dual-polarization radar at Gadanki, showing embedded convective cells in the Z field (color) around Gadanki. (b) Temporal variation of IWV on July 13 (solid line) and climatological GNSS-derived IWV for July (dashed line), depicting the increase in IWV on a storm day relative to the climatology. Histograms represent 30-min accumulated rainfall. (c) Temporal variation of meteorological parameters, temperature, and wind speed obtained with an AWS and INSAT-derived T_b.

be seen at the time of rainfall. The GNSS-derived IWV started increasing at 12 UTC, reached a maximum at the time of peak rainfall, and decreased rapidly after the rainfall. The dramatic increase in IWV is much more than the climatological daily diurnal variations (dashed line), indicating that the increase is associated with the storm. The increase in IWV started ∼4 h before the rain (storm) occurrence at Gadanki and the increase is about 8 mm. It is clear from Fig. 2(b) that the IWV increase is an excellent predictor for the occurrence of storm, as the increase is seen well ahead of the storm.

In addition to the above parameters, Benevides et al. [18] suggested the use of gradient in T_b to reduce the false alarm rate. Dramatic change in T_b can be clearly seen [see Fig. 2(b)] prior to and during the occurrence of storm/rainfall. The T_b value has reduced by ∼90 K in 3 h. Even prior to the storm, the reduction in T_b was significant, indicating that it can also act as good predictor. Fig. 2(b) clearly shows the signature of storm in various surface meteorological parameters and T_b, i.e., reduction in temperature and T_b, increase in wind speed, high IWV, and temporal changes in IWV. To identify and predict the storms, one should use proper thresholds to the above parameters. Larger thresholds will only identify intense storms, while smaller thresholds will increase false alarm rates. In other words, different thresholds provide different prediction outcomes; therefore, determining an ideal threshold value for high prediction capacity and least false alarm rate is essential. Data from September 2018 to December 2019 are used to identify storms and the data corresponding to chosen predictors during the passage of these storms are used to generate the threshold values for predictors. The number of storms is large during the monsoon season (65), allowing to obtain robust thresholds for predictors. It includes all types of convective storms/cells, including 33 mild ($40 \leq Z < 45$ dBZ), 17 intense ($45 \leq Z < 50$ dBZ), and 15 extreme ($Z > 50$ dBZ) storms. Winter and premonsoon seasons have been not considered as there were not enough storm occurrences to meaningfully derive thresholds for predictors.

The histograms of the maximum value of IWV, temporal gradients of IWV, temperature and wind speed and T_b, and the time at which the IWV starts to increase (additional IWV) relative to the time of storm (rainfall) (Δt) occurrence are plotted using the data from storm days, identified as per the procedure outlined above. The magnitude of gradients in temperature is found to be different for day and night, smaller during the night (peak is at 3 K) than in the day (peak is at 5 K). However, it is found from many case studies that the rapid change in temperature and wind speed occurred at the time of event, while the gradients are found to be weaker prior to the storm/rainfall occurrence. They may be good to do nowcasting with a lead time of 30 min but may not be suitable for longer lead times. The histograms of remaining parameters, magnitude of IWV, temporal gradients in IWV, and T_b and Δt are shown in Fig. 3.

The IWV is generally high during the monsoon season and additional IWV associated with storms increases the IWV further. The histogram of maximum IWV on storm days shows values larger than 50 mm with a peak in the range of 58–62 mm [see Fig. 3(a)]. Not only IWV values, but the rate of increase of IWV is also high on the storm days with all the cases showing a rate $\geq$ 1.54 mm·hr^{-1}. The rate of IWV values as large as 5–9 mm·hr^{-1} was also observed; nevertheless, their occurrence is found to be less [see Fig. 3(b)]. However, the histogram shows that on 50% of the days, and the rate of IWV values is nearly equal to 2 mm·hr^{-1} (1–5–2.5-mm·hr^{-1}·bin). The storms can also be recognized by the rapid growth of clouds. The histogram of temporal change in T_b shows values $\geq$ 5 K·hr^{-1} [see Fig. 3(c)]. However, the cloud growth is much larger in many of the cases (∼70% occurrence). From Fig. 2(d), it is evident that in most of the cases, IWV enhancement can typically be seen 2 h before a storm, indicating that the majority of the

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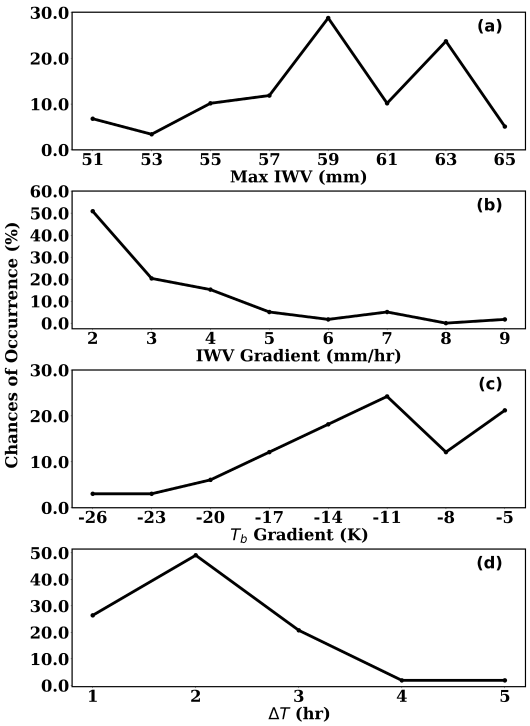


Fig. 3. Histograms of (a) maximum IWV, (b) IWV gradient, (c) T_b gradient, and (d) time difference between the time of rain occurrence and time at which IWV rapidly increases, generated using observations from 65 stormy days.

cases can be identified 2 h before the storm. However, some storms show an increase in IWV 1 h before the occurrence of storm/rainfall. These details, particularly the lowest values in the histograms (representatives of weakest storms), are used to arrive at thresholds to identify the storms. Therefore, the thresholds of IWV, rate of IWV, and rate of T_b are finalized as 50 mm, 1.56 mm hr^{-1}, and 5 K km^{-1}, respectively.

The above thresholds are compared and contrasted with those available in the literature at different geographical locations. Except for T_b, which was used only by Benevides et al. [18], other parameters are extensively used by several researchers for identifying/predicting storms/rainfall [4], [5], [13], [14]. Yao et al. [5] examined several rain events in Zhejiang province in the subtropical monsoon-affected zone in China and noted maximum IWV and rate of IWV are in the range of 51.1–73.9 mm and 0.93–7 mm·hr^{-1}, respectively. The threshold IWV value used in the present study, i.e., 50 mm of maximum IWV, is very close to the lowest range noted by Yao et al. [5]. Also, the rate of IWV in the present study is within the range observed by Yao et al. [5]. Sapucci et al. [13] also reported a rate of 2 mm·hr^{-1} IWV before intense rainfall in CHUVA vale campaign in Brazil. Furthermore, they noted that the IWV peaks 16–96 min before the maximum rainfall and used the average rate of IWV to nowcast storms. Benevides et al. [4] also noted the change in IWV around 1.4 mm·hr^{-1}, which is very close to the value used in the present study, for intense rain events in Portugal. However, some of the earlier studies have used much smaller thresholds for IWV gradient (e.g., 0.6 by Yao et al. [5]) for identifying light rain. Zhao et al. [14] also used a smaller threshold for IWV gradient for their rainfall detection algorithm in China. All these studies considered the increase in IWV from smallest to largest values of IWV before the storm and then normalized by the time to obtain hourly values. As shown in Fig. 2(b), the minimum to maximum IWV includes the diurnal variability, which is a slowly varying component and increases the time between the minimum and maximum IWV, thereby reducing the IWV gradient. However, IWV above the climatological diurnal variation is considered to be associated with the storm in the present study. This increase in IWV is quite rapid, and the gradient is also large as seen in the case study. Also, given the large diurnal amplitudes of IWV (of the order of > 2 mm), smaller gradients may increase the false alarm. Therefore, a larger gradient in IWV is used in the present study.

Considering the large seasonal variability in IWV, earlier studies suggested the use of different IWV thresholds for different seasons for identifying storms [5], [14]. Given such constraints, the thresholds derived in the present study are valid only for the monsoon season, and for the other seasons, the thresholds need to be fine-tuned.

IV. Validation and Discussion

A. Evaluation Metrics

To examine the prediction outcome, the following metrics are estimated [29]:

$$\text{Probability of Detection} = \frac{d}{d+\text{c}} * 100 \tag{4}$$

$$\text{False Alarm Rate} = \frac{b}{b+d} * 100 \tag{5}$$

$$\text{Miss Rate} = \frac{c}{c+d} * 100 \tag{6}$$

$$\text{Model Accuracy} = \frac{a+d}{a+b+c+d} * 100 \tag{7}$$

where a is the total number of truly detected nonstorm events in accordance with the actual data by the algorithm and b is the number of incorrectly recognized storm occurrences. The predicted number of nonstorm events, when a storm was really occurring, is represented by c, which stands for the events that the algorithm missed. True detection of storm events is represented by d.

B. Validation and Discussion

The validation of storm prediction using the above two-step process has been carried out using observations from the year 2020. A total of 54 storms (including 17, 13, and 24 mild, intense, and extreme storms, respectively) have been identified during the monsoon season of 2020 (validation period) using the procedure outlined in Section III-B, i.e., from X-band radar reflectivity, decrease in temperature, increase in wind speed, IWV, and change in $T_b = 5$ K·hr^{-1} and considered them as reference dataset. The procedure outlined in Section III-B, i.e., first predicting the IWV using light GBM and then using the thresholds for predicted IWV magnitude and variation and INSAT-derived T_b, has been followed to nowcast storms. The nowcasted storms are tested against the reference dataset to validate the algorithm and thresholds.

The algorithm's sensitivity on the use of one predictor or multiple predictors has been tested by comparing with

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TABLE II
EVALUATION STATISTICS OF SINGLE-, DOUBLE-, AND THREE-PARAMETER SCHEMES IN PREDICTING STORMS ALONG WITH THRESHOLD VALUES USED FOR EVALUATION

	Threshold Values	Probability of detection (%)	Miss Rate (%)	False Alarm (%)	Model Accuracy (%)
IWV (mm)	> 50	100	0	53.4	49.18
IWV Rate (mm/hr)	> 1.56	100	0	48.0	58.19
T_b Rate (K/hr)	< -5	100	0	55.00	45.90
IWV, IWV Rate	>50 , >1.56	100	0	47.50	59.80
IWV, IWV Rate, T_b Rate	>50, >1.56, <-5	100	0	5.20	97.50
Cape(J/kg), K Index (K)	>500, >30	87.04	12.96	33.80	74.59
Cape(J/kg), K Index (K), PW, PW Rate	>500, >30, >50, >1.56	87.04	12.96	11.32	89.34

reference dataset, and the evaluation statistics are shown in Table II. Since the smallest values of histograms were taken as thresholds for the respective parameters, the model is able to detect all the storms/rain events with single or multiple parameters (except for stability indices, such as CAPE and K-index), as evidenced by 100% probability of detection. However, the false alarm rate is high (47%–55%), when the model uses only single parameter (IWV/IWV gradient/Tb gradient) or two parameters (IWV and IWV rate) for nowcasting. For instance, the IWV exceeds 50 mm on many days as we are taking data from monsoon season, during which the IWV is generally high. This increases false alarm rate alarmingly to 53%. Similarly, the usage of smallest gradient for T_b gradient increases the false alarm rate to 55%. One can notice high false alarm rate of 48% even with a larger threshold of IWV gradient than those available in the literature. The usage of smaller threshold will drastically increase the false alarm rate and reduces the model accuracy.

Traditionally meteorologists employ stability indices, such as convective available potential energy (CAPE), K-index, and so on, estimated from vertical soundings of temperature and humidity, for predicting storms [3]. The above parameters are estimated using the fifth-generation ECMWF atmospheric reanalysis of the global climate (ERA 5) data 30, 60, 90, and 120 m before the storm/rainfall occurrence to capture the instability during prestorm environment. Similar to Fig. 3, histograms have been constructed, and the smallest values are considered as thresholds for nowcasting. Based on histograms, the thresholds for CAPE and K-index are finalized as 500 J·Kg^{-1} and 30 K, respectively. The 30-K threshold agrees with many earlier studies in India, and the 500-J·Kg^{-1} threshold for CAPE is smaller than the reported values, which are typically found to be larger than 1000 J·Kg^{-1}.

However, these storms are mostly intense in nature. The inclusion of all storms in the present study, including weak storms, reduced the threshold. It is also possible that ERA5 may not be able to properly represent the prestorm environment for all storms. Nevertheless, the usage of these two parameters certainly reduced the false alarms compared to single- and double-parameter models based on IWV or T_b. Still, the model missed some of the storms.

TABLE III
SENSITIVITY OF THREE-PARAMETER SCHEME IN THE PREDICTION OF STORMS WITH DIFFERENT LEAD TIMES

	Probability of detection (%)	Miss Rate (%)	False Alarm (%)	Model Accuracy (%)
30 min	100	0	6.89	96.58
60 min	96.29	3.70	8.77	94.01
90 min	92.59	7.40	10.71	91.45
120 min	87.50	12.50	12.50	88.00

Multiparameter models are found to be working well in nowcasting storms. For instance, the introduction of the rate of change in brightness temperature as a third parameter in addition to the IWV and IWV gradient parameters has resulted in a noticeable improvement in the prediction algorithm. The three-parameter model lowered the false alarm rate drastically to 5.2% and increased the model accuracy to 97.50%. Similarly, the inclusion of IWV and IWV parameters in CAPE and K-index-based model reduces the false alarm rate to 11% and increases model accuracy to 89%. It is clear from Table II that the three-parameter model using IWV, IWV gradient, and T_b gradient works well with 100% detection capability of storms with a smaller false alarm rate.

The above analysis is carried out with 30-min predicted IWV. To examine the sensitivity of predicted IWV with different lead times on storm nowcasting, the IWV predicted at different time intervals (30, 60, 90, and 120 min) has been (like in Fig. 2) used in the model. Three-parameter models with IWV, IWV gradient, and T_b gradient as inputs are used for sensitivity analysis. It can be seen from Table III that no storms were missed with 30-min lead time (detection rate = 100%), but about 12% of total storms were not detected with 120-min lead time. The algorithm's performance degrades gradually with the increase in lead time. It can be seen that the probability of detection decreases gradually from 100% to 87.5%, and missing rate and false alarms increase to ∼7%–12.5%, respectively, as lead time increases from

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30 to 120 min. Nevertheless, the models' accuracy remains above 88% even with 120-min lead time, which is comparable or better than the existing methods/models in the literature.

V. Conclusion

The GNSS-based storm prediction techniques, which are increasingly popular in recent years, essentially follow one of the following approaches: AI/ML techniques or threshold-based schemes. In the present study, a new hybrid scheme is introduced based on the strengths of both approaches. The algorithm first predicts the IWV using a machine learning technique (light GBM) with GNSS and surface measurements as inputs and then nowcast storms based on multiparameter thresholding algorithm. The accuracy of predicted IWV with 2-h lead time is ∼2 mm and the accuracy is found to be much better with shorter lead times (30–90 min). The main advantage of the present scheme is that it uses predicted IWV to nowcast storms, which provides more time for disaster preparedness and rescue operations.

The thresholds employed in the present algorithm are purely based on 18 months of GNSS-based IWV measurements made during the stormy days (from the years 2018 and 2019), which are identified based on radar and AWS observations. The threshold for the magnitude of IWV agrees quite well with those available in the literature at different geographical locations, indicating the robustness of this threshold. The study also clearly shows the increase in IWV starts few hours (1–4 h) before the storm/rainfall occurrence, indicating the predictive potential of IWV gradient. However, the IWV gradient used in the present study (1.56 mm·hr^{-1}) is larger than that used in some of the earlier studies. The smaller thresholds for IWV gradient in earlier studies are mainly due to the inclusion of routine diurnal variation in the estimation of gradient. In the present study, climatological monthly diurnal variation of IWV has been used to identify storm associated IWV gradient and the threshold for gradient IWV is chosen from the storm associated IWV gradient histogram. Smaller thresholds of IWV gradient will increase false alarm rates dramatically as reported in earlier studies.

The evaluation of the model has been carried out by comparing its outcome with a reference dataset of 54 stormy days in the year 2020 that were identified with an X-band radar and AWS. The sensitivity of the individual predictors and combination of predictors in nowcasting storms have also been studied. In addition, the efficacy of traditionally used stability indices for storm prediction has tested. Because of the smaller thresholds used in the present study, the probability of detection of storms is found to be 100% with IWV magnitude, IWV gradient, T_b gradient, and any combination of these parameters. However, the false alarm rate is startlingly high with the usage of single or two predictors for nowcasting purpose. The inclusion of T_b gradient to IWV magnitude and gradients drastically reduced the false alarm rates to ∼5% and increased the success ratio to 97.5%. The stability parameters, such as CAPE and K-index, failed to predict storms accurately with 87% probability of detection and ∼34% false alarm rate. The addition of IWV parameters improved the accuracy of the model by ∼15%. The study unequivocally demonstrates the importance of IWV in nowcasting storms as it improves the model's success ratio with both traditional stability indices and T_b gradient.

The ability of the three-parameter model (IWV magnitude and gradient and T_b gradient) in nowcasting with different lead times has also been tested. It is noted, in general, that the model performance degrades with the increase in lead time with the increase in the percentage of missing storms and false alarm rates. However, the model's accuracy still remains close to 90% with 120-min lead time, which is much better than the existing methods.

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